

CSE 676 Deep Learning Group Project

Project Topic - Fake News Detection

Our project focuses on **Fake News Detection**, which is the task of automatically classifying news content as **real** or **fake** using Natural Language Processing (NLP) and deep learning techniques. Fake news has become a major issue due to the rapid spread of misinformation through online platforms. Detecting fake news using machine learning models can help improve information reliability and reduce the spread of misleading content.

Dataset Selection

We plan to use publicly available datasets for fake news classification:

- **LIAR dataset** – contains short political statements labeled as true or false
- **FakeNewsNet dataset** – contains longer news articles collected from online sources

For the initial phase of the project, we will start with the **LIAR dataset** because it is structured and easier to preprocess. Later, we may extend our work to **FakeNewsNet** to evaluate model performance on longer news articles.

Dataset links

LIAR Dataset

- Paper page: [LIAR dataset \(UW\)](#)

FakeNewsNet

- GitHub: [FakeNewsNet repository](#)

Progress So Far

We have completed the following steps:

- Selected **Fake News Detection** as our project topic
- Reviewed project requirements and identified suitable datasets
- Explored the structure of the LIAR dataset
- Defined the problem as a **binary text classification task**
- Planned preprocessing steps including text cleaning, tokenization, and label encoding
- Planned to divide the dataset into **training, validation, and testing sets**
- Considered possible challenges such as **class imbalance** and **data leakage**
- Identified baseline and deep learning models suitable for this problem

Planned Methodology

Data Preprocessing

We plan to perform the following preprocessing steps:

- Convert text to lowercase
- Remove unnecessary symbols or formatting if required
- Tokenization of text data
- Padding or truncating sequences to fixed length
- Encoding labels into numerical format
- Splitting data into train, validation, and test sets

Models to be Implemented

We will implement multiple models to compare performance:

Baseline Model

- TF-IDF with Logistic Regression

Deep Learning Models

- LSTM or BiLSTM model for sequence learning
- BERT or DistilBERT transformer model for contextual text understanding

These models will allow us to compare traditional machine learning approaches with deep learning approaches.

Evaluation Metrics

We will evaluate model performance using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Since fake news datasets may have imbalanced classes, F1-score will be an important metric.

Comparison and Analysis Plan

We plan to compare:

- Traditional machine learning vs deep learning models
- LSTM vs transformer-based models
- Impact of preprocessing techniques on performance

We will also perform **error analysis** to identify cases where the model makes incorrect predictions. For example:

- statements that are ambiguous
- satire or opinion-based content
- politically sensitive statements
- very short or very long text inputs

Ethical Considerations and Limitations

Fake news detection models may sometimes misclassify true information as false or vice versa. Bias in training data may affect model predictions. We will discuss these limitations and possible improvements in the final report.